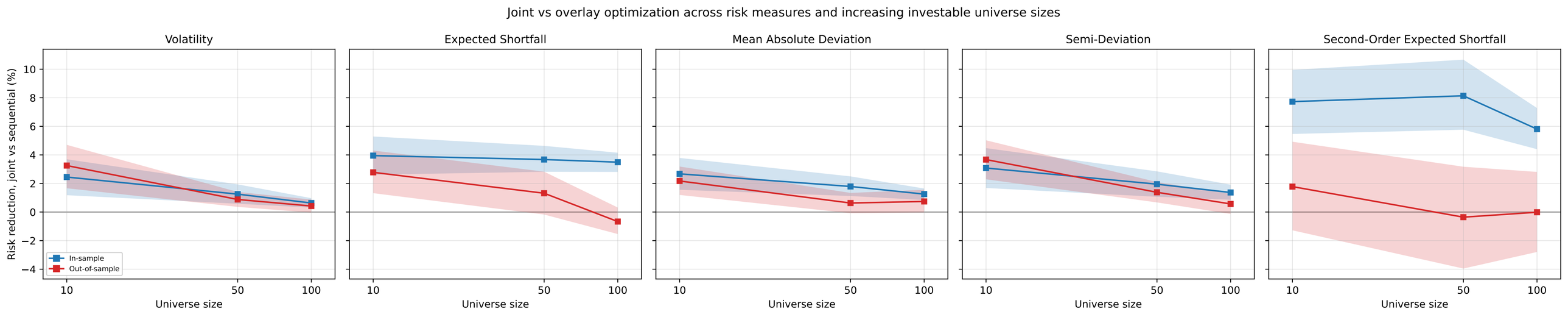
Joint vs overlay optimization across risk measures and increasing investable universe sizes
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Second-Order Expected Shortfall
Risk reduction, joint vs sequential (%)
10
8
6
4
2
0
−2
−4
In-sample
Out-of-sample
10
50
100
Universe size